

5th Exercise Sheet

TUTORIUM: Friday, 12.06.2026.

SECTION

010

Van Hove singularities

PROBLEM

Consider the dispersion relation (single-particle energy states) for electrons on a simple hypercubic lattice in d dimensions, with only nearest-neighbor hopping:

$$\varepsilon_{\mathbf{k}} = -2t \sum_{i=1}^d \cos k_i, \quad (1)$$

with the hopping amplitude t and the lattice constant $a = 1$. The density of single-particle states in this system is then given by

$$N(\varepsilon) = \frac{1}{(2\pi)^d} \int_{-\pi}^{\pi} d^d k \delta(\varepsilon - \varepsilon_{\mathbf{k}}). \quad (2)$$

In the first exercise you have calculated numerically and then plotted these densities of states for $d = 1, 2, 3$. Here, the singular structures (divergences, cusps) of these functions should be analyzed analytically.

a)

PROBLEM

Calculate $N(\varepsilon)$ for $d = 1$ explicitly and determine the interval $[\varepsilon_1, \varepsilon_2]$ on which $N(\varepsilon) \neq 0$. Moreover, identify the values ε^* where D diverges, i.e. where $N(\varepsilon^*) = \infty$. From which points $\mathbf{k}^*$ in the dispersion relation originate these divergences? Show that the divergences can be reproduced by taking into account only the contributions from these $\mathbf{k}^*$ -points.

Hint: Replace $\varepsilon_{\mathbf{k}}$ in Equation 2 by a corresponding Taylor-expansion around these points up to second order.

SOLUTION

From Exercise 1, for $d = 1$ the dispersion is

$$\varepsilon_k = -2t \cos(k), \quad (3)$$

so the density of states is

$$N(\varepsilon) = \begin{cases} \frac{1}{\pi\sqrt{4t^2-\varepsilon^2}} & \text{if } |\varepsilon| < 2t \\ 0 & \text{if } |\varepsilon| > 2t \end{cases}. \quad (4)$$

Thus the band is supported on

$$\varepsilon \in [-2t, 2t] \quad (5)$$

and the DOS diverges at the band edges

$$\begin{aligned} \varepsilon_1^* &= -2t \\ \varepsilon_2^* &= +2t \end{aligned} \quad (6)$$

The dispersion has value $\varepsilon_1^* = -2t$ at

$$\mathbf{k}_1^* = 0 \quad (7)$$

and $\varepsilon_2^* = +2t$ at

$$\mathbf{k}_2^* = \pm\pi. \quad (8)$$

- Expanding $\varepsilon_{\mathbf{k}}$ around $\mathbf{k}_1^* = 0$ (a minimum, with $\varepsilon_{\mathbf{k}_1^*} = -2t$) gives

$$\varepsilon_{\mathbf{k}} = -2t + k^2 t + \mathcal{O}(k^4) \quad (9)$$

which yields for the DOS

$$N_1(\varepsilon) = \begin{cases} 0 & \text{if } \varepsilon < -2t \\ \frac{1}{2\pi\sqrt{t(2t+\varepsilon)}} & \text{if } -2t \leq \varepsilon \end{cases}, \quad (10)$$

a band-edge singularity at the onset ($\varepsilon = -2t$) of DOS.

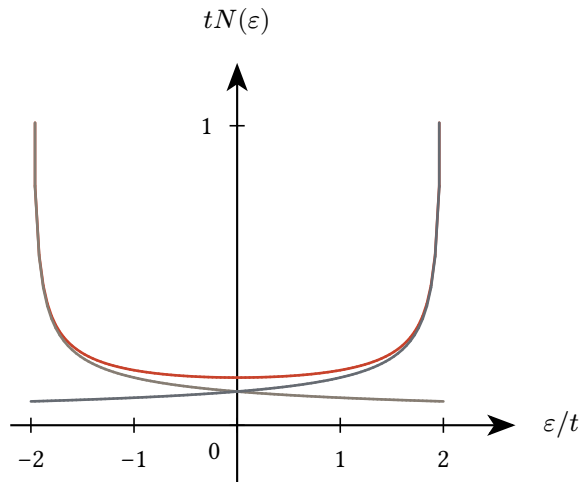
- Expanding $\varepsilon_{\mathbf{k}}$ around $\mathbf{k}_2^* = \pm\pi$ (a maximum with $\varepsilon_{\mathbf{k}_2^*} = 2t$) gives

$$\varepsilon_{\mathbf{k}} = 2t - (k \mp \pi)^2 t + \mathcal{O}((k \mp \pi)^4) \quad (11)$$

which yields for the DOS

$$N_2(\varepsilon) = \begin{cases} \frac{1}{2\pi\sqrt{t(2t-\varepsilon)}} & \text{if } \varepsilon \leq 2t \\ 0 & \text{if } 2t < \varepsilon \end{cases}, \quad (12)$$

a band-edge singularity at the cutoff ($\varepsilon = +2t$) of DOS.



The plot shows that $N_{\pm}(\varepsilon)$ (grey curves) exhibit the same divergences as $N(\varepsilon)$.

b)

PROBLEM

For $d = 2$ one can show that $N(\varepsilon)$ is essentially given by a complete elliptic integral of the first kind. Here, however, only the singular contributions to $N(\varepsilon)$ should be analyzed. As in the one-dimensional case a singular contribution originates from stationary points in the dispersion relation. Determine the kind of stationary point (i.e., maximum, minimum or saddle point) which generates this so-called Van Hove singularity in the two-dimensional DOS and determine the singular contribution to $N(\varepsilon)$ by expanding $\varepsilon_{\mathbf{k}}$ around the corresponding stationary point in Equation 2 as for the one-dimensional case above.

SOLUTION

The dispersion relation has gradient

$$\nabla \varepsilon_{\mathbf{k}} = 2t \begin{pmatrix} \sin k_x \\ \sin k_y \end{pmatrix} \quad (13)$$

and thus stationary points at

$$\mathbf{k}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (14)$$

(minimum),

$$\mathbf{k}_2 = \begin{pmatrix} \pi \\ \pi \end{pmatrix} \quad (15)$$

(maximum) and

$$\mathbf{k}_3 = \begin{pmatrix} 0 \\ \pi \end{pmatrix} \quad (16)$$

(saddle point).

Note 1: π should be taken to mean $\pm\pi$ (periodic BC).

Note 2: There is also $\begin{pmatrix} \pi \\ 0 \end{pmatrix}$, which is equivalent by lattice symmetry to $\mathbf{k}_3$.

- We first expand around $\mathbf{k}_1$ where $\varepsilon_{\mathbf{k}_1} = -4t$:

$$\varepsilon_{\mathbf{k}} = -4t + \|\mathbf{k}\|^2 t + \mathcal{O}(\mathbf{k}^4) \quad (17)$$

which yields for the DOS

$$N_1(\varepsilon) = \begin{cases} 0 & \text{if } \varepsilon < -4t \\ \frac{1}{4\pi t} & \text{if } -4t \leq \varepsilon \end{cases} \quad (18)$$

i.e. a band-edge singularity at DOS onset.

- We now expand around $\mathbf{k}_2$ where $\varepsilon_{\mathbf{k}_2} = 4t$:

$$\varepsilon_{\mathbf{k}} = 4t - \|\mathbf{k} - \mathbf{k}_2\|^2 t + \mathcal{O}((\mathbf{k} - \mathbf{k}_2)^4) \quad (19)$$

which yields for the DOS

$$N_2(\varepsilon) = \begin{cases} \frac{1}{4\pi t} & \text{if } \varepsilon \leq 4t \\ 0 & \text{if } 4t < \varepsilon \end{cases}, \quad (20)$$

i.e. a band-edge singularity at DOS cutoff.

- Lastly, we treat the saddle-point $\mathbf{k}_3$ where $\varepsilon_{\mathbf{k}_3} = 0$:

$$\varepsilon_{\mathbf{k}} = k_x^2 t - (k_y - \pi)^2 t + \mathcal{O}((\mathbf{k} - \mathbf{k}_3)^4) \quad (21)$$

which yields for the DOS

$$N_3(\varepsilon) = \frac{1}{2\pi^2 t} \operatorname{arcosh} \left(\pi \sqrt{\frac{t}{|\varepsilon|}} \right) \quad (22)$$

i.e. a logarithmic singularity at $\varepsilon = 0$. Around $\varepsilon \rightarrow 0$, it has the leading asymptotic form

$$N_3(\varepsilon) \rightarrow \frac{1}{2\pi^2 t} \log \left(2\pi \sqrt{\frac{t}{|\varepsilon|}} \right) = -\frac{1}{4\pi^2 t} \log |\varepsilon| + \frac{1}{4\pi^2 t} \log(4\pi^2 t). \quad (23)$$

The contribution to the DOS has been doubled here because of the aforementioned second saddle point.

From Exercise 1, for $t = 1$ the two-dimensional square-lattice DOS was

$$N(\varepsilon) = \begin{cases} \frac{1}{2\pi^2 t} K \left(1 - \frac{\varepsilon^2}{16t^2} \right) & \text{if } |\varepsilon| < 4t \\ 0 & \text{if } |\varepsilon| > 4t \end{cases}, \quad (24)$$

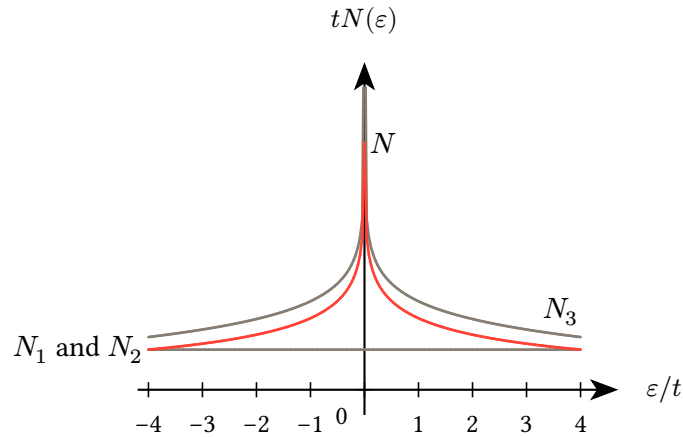
where

$$K(m) = \int_0^{\frac{\pi}{2}} \frac{d\varphi}{\sqrt{1 - m \sin^2(\varphi)}}. \quad (25)$$

Thus the singular feature already found in Exercise 1 is a logarithmic Van Hove singularity at

$$\varepsilon = 0, \quad (26)$$

while the band edges at $\varepsilon = \pm 4t$ have finite DOS.



c)

PROBLEM

Try to predict how the singular behavior of the DOS evolves with the dimensions of the system for $d \geq 3$.

SOLUTION

We have that:

- Minimum of $\varepsilon_{\mathbf{k}} \Rightarrow$ Singularity at DOS onset
- Maximum of $\varepsilon_{\mathbf{k}} \Rightarrow$ Singularity at DOS cutoff
- Saddle point of $\varepsilon_{\mathbf{k}} \Rightarrow$ Singularity within DOS

So, for $d \geq 3$:

- Minimum at $\mathbf{k} = 0$:

DOS “starts” at $\varepsilon = -2dt$ with a band-edge nonanalyticity.

- Maximum at $\mathbf{k} = (\pi, \pi, \dots, \pi)^T$:

DOS “ends” at $\varepsilon = +2dt$ with a band-edge nonanalyticity.

- Saddle points at $\mathbf{k} = (\underbrace{0, \dots, 0}_p, \underbrace{\pi, \dots, \pi}_{d-p})^T$ with $1 \leq p < d$:

DOS has a singularity at $\varepsilon = 2(d - 2p)t$.

Examples:

- $d = 3$

Start at $-6t$, singularities at $-2t$ and $2t$, end at $6t$.

- $d = 4$

Start at $-8t$, singularities at $-4t$, 0 and $4t$, end at $8t$.

d)

PROBLEM

(Bonus points) Finally, consider the limit $d \rightarrow \infty$. In this case, one has to rescale the hopping amplitude as $t \rightarrow t/\sqrt{d}$ in order to render the total energy of the system as well as the second moment (standard deviation) of the density of state finite. Show that $N_\infty(\varepsilon)$ is proportional to a Gauß distribution.

SOLUTION

After applying the rescaling, we have

$$\varepsilon_{\mathbf{k}} = -\frac{2t}{\sqrt{d}} \sum_{i=1}^d \cos k_i \quad (27)$$

and we will employ the relation

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} ds e^{ixs}, \quad (28)$$

which yields for the DOS in d dimensions

$$\begin{aligned} N(\varepsilon) &= \frac{1}{(2\pi)^d} \int_{-\pi}^{\pi} d^d k \delta\left(\varepsilon + \frac{2t}{\sqrt{d}} \sum_{i=1}^d \cos k_i\right) \\ &= \frac{1}{(2\pi)^d} \int_{-\pi}^{\pi} d^d k \frac{1}{2\pi} \int_{-\infty}^{\infty} ds \exp\left(i\left(\varepsilon + \frac{2t}{\sqrt{d}} \sum_{i=1}^d \cos k_i\right)s\right) \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} ds e^{i\varepsilon s} \left[\underbrace{\frac{1}{2\pi} \int_{-\pi}^{\pi} dk \exp\left(i \frac{2st}{\sqrt{d}} \cos k\right)}_{J_0\left(\frac{2st}{\sqrt{d}}\right) \text{ Bessel function of the first kind}} \right]^d \\ &\stackrel{d \rightarrow \infty}{=} \frac{1}{2\pi} \int_{-\infty}^{\infty} ds e^{i\varepsilon s} e^{-s^2 t^2} \\ &= \frac{1}{2t\sqrt{\pi}} e^{-\varepsilon^2/4t^2}. \end{aligned} \quad (29)$$

Magnetic susceptibilities in d dimensions

Consider a system of non-interacting electrons on a (hyper)cubic lattice whose energy dispersion is given by Equation 1.

a)

Compute the magnetic susceptibility, i.e. the Fourier transform of the spin-spin response function

$$\langle T_{\tau} S^z(\mathbf{r}_i, \tau) S^z(0, 0) \rangle, \quad (30)$$

with

$$S^z(\mathbf{r}_i, \tau) = \frac{\hbar}{2} [c_{i\uparrow}^{\dagger}(\tau) c_{i\uparrow}(\tau) - c_{i\downarrow}^{\dagger}(\tau) c_{i\downarrow}(\tau)], \quad (31)$$

for the frequency $\Omega_m = 0$ (static susceptibility), and for the two momenta $\mathbf{Q} = (0, 0, 0, \dots)$ (ferromagnetic susceptibility) and $\mathbf{Q} = (\pi, \pi, \pi, \dots)$ (antiferromagnetic susceptibility).

We combine $(\mathbf{r}_i, \tau) \equiv r$ and apply Wick's theorem (the system is non-interacting)

$$\begin{aligned}\chi(r) &= \langle T S^z(r) S^z(0) \rangle \\ &= \frac{\hbar^2}{4} \langle T (c_{\uparrow}^{\dagger}(r) c_{\uparrow}(r) - c_{\downarrow}^{\dagger}(r) c_{\downarrow}(r)) (c_{\uparrow}^{\dagger}(0) c_{\uparrow}(0) - c_{\downarrow}^{\dagger}(0) c_{\downarrow}(0)) \rangle \\ &= \frac{\hbar^2}{4} \sum_{\sigma} \langle T c_{\sigma}^{\dagger}(r) c_{\sigma}(0) \rangle \langle T c_{\sigma}(r) c_{\sigma}^{\dagger}(0) \rangle\end{aligned}\quad (32)$$

We can now substitute the Green's function

$$G_{\sigma}(r) = -\langle T c_{\sigma}(r) c_{\sigma}^{\dagger}(0) \rangle \quad (33)$$

to get

$$\chi(r) = -\frac{\hbar^2}{4} \sum_{\sigma} G_{\sigma}(-r) G_{\sigma}(r) = -\frac{\hbar^2}{2} G(-r) G(r). \quad (34)$$

Fourier transform to get the susceptibility

$$\begin{aligned}\chi(\mathbf{q}, i\Omega_m) &= \sum_i \int_0^{\beta} d\tau \chi(\mathbf{r}_i, \tau) e^{i\mathbf{q} \cdot \mathbf{r}_i + i\Omega_m \tau} \\ &= -\frac{\hbar^2}{2} \sum_i \int_0^{\beta} d\tau G(-\mathbf{r}_i, -\tau) G(\mathbf{r}_i, \tau) e^{i\mathbf{q} \cdot \mathbf{r}_i + i\Omega_m \tau} \\ &= -\frac{\hbar^2}{2} \frac{1}{N\beta} \sum_{\mathbf{k}} \sum_{i\omega_n} G(\mathbf{k}, i\omega_n) G(\mathbf{k} + \mathbf{q}, i\omega_n + i\Omega_m)\end{aligned}\quad (35)$$

with the non-interacting Green's function

$$G(\mathbf{k}, i\omega_n) = \frac{1}{i\omega_n - \varepsilon_{\mathbf{k}}}. \quad (36)$$

Ferromagnetic susceptibility (at $\mathbf{0}$):

$$\begin{aligned}\chi_F = \chi(\mathbf{0}, 0) &= -\frac{\hbar^2}{2} \frac{1}{N\beta} \sum_{\mathbf{k}} \sum_{i\omega_n} G(\mathbf{k}, i\omega_n) G(\mathbf{k}, i\omega_n) \\ &= -\frac{\hbar^2}{2} \frac{1}{N} \sum_{\mathbf{k}} n'_F(\varepsilon_{\mathbf{k}})\end{aligned}\quad (37)$$

Antiferromagnetic susceptibility (at π):

We work at half filling, i.e. $\mu = 0$, so that

$$\varepsilon_{\mathbf{k}+\pi} = -\varepsilon_{\mathbf{k}}. \quad (38)$$

and then the susceptibility is

$$\begin{aligned}
\chi_{\text{AF}} = \chi(\boldsymbol{\pi}, 0) &= -\frac{\hbar^2}{2} \frac{1}{N\beta} \sum_{\mathbf{k}} \sum_{i\omega_n} G(\mathbf{k}, i\omega_n) G(\mathbf{k} + \boldsymbol{\pi}, i\omega_n) \\
&= -\frac{\hbar^2}{2} \frac{1}{N} \sum_{\mathbf{k}} \frac{n_{\text{F}}(\varepsilon_{\mathbf{k}}) - n_{\text{F}}(\varepsilon_{\mathbf{k}+\boldsymbol{\pi}})}{\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}+\boldsymbol{\pi}}} \\
&= -\frac{\hbar^2}{2} \frac{1}{N} \sum_{\mathbf{k}} \frac{n_{\text{F}}(\varepsilon_{\mathbf{k}}) - n_{\text{F}}(-\varepsilon_{\mathbf{k}})}{2\varepsilon_{\mathbf{k}}} \\
&= \frac{\hbar^2}{4} \frac{1}{N} \sum_{\mathbf{k}} \frac{1}{\varepsilon_{\mathbf{k}}} \tanh\left(\frac{\beta\varepsilon_{\mathbf{k}}}{2}\right)
\end{aligned} \tag{39}$$

b)

PROBLEM

Determine the leading divergences of the ferromagnetic and the antiferromagnetic susceptibilities for $T \rightarrow 0$ in $d = 2$ dimensions. To this end write the total density of states as a sum of a singular and a regular contribution as calculated in Problem 10b).

- Hint: It is convenient to write the total density of states as $N(\varepsilon) = N_{\text{reg}}(\varepsilon) + N_{\text{sing}}(\varepsilon)$, and treat the two contributions separately. For the ferromagnetic case, expand N_{reg} around zero (Sommerfeld-like expansion) and, for the singular part, change variable to $x = \beta\varepsilon$ to make the temperature dependence manifest. For the antiferromagnetic case, (i) differentiate the susceptibility with respect to β before integrating, (ii) proceed as in the ferromagnetic case, (iii) integrate back over β at the end.

SOLUTION

We write

$$N(\varepsilon) = N_{\text{reg}}(\varepsilon) + N_{\text{sing}}(\varepsilon) \tag{40}$$

with

$$N_{\text{sing}}(\varepsilon) = A \log|\varepsilon| \tag{41}$$

where

$$A = -\frac{1}{4\pi^2 t}. \tag{42}$$

We also rewrite the susceptibilities as

$$\begin{aligned}
\chi_{\text{F}} &= -\frac{\hbar^2}{2} \int d\varepsilon N(\varepsilon) n'_{\text{F}}(\varepsilon) \\
\chi_{\text{AF}} &= \frac{\hbar^2}{4} \int d\varepsilon N(\varepsilon) \frac{1}{\varepsilon} \tanh\left(\frac{\beta\varepsilon}{2}\right).
\end{aligned} \tag{43}$$

- Ferromagnetic: Write

$$n'_{\text{F}}(\varepsilon) = -\frac{\beta}{4 \cosh^2(\beta\varepsilon/2)} \tag{44}$$

and then, with $x = \beta\varepsilon$,

$$\begin{aligned}
\chi_F &\sim -\frac{\hbar^2}{2} \int d\varepsilon N_{\text{sing}}(\varepsilon) n'_F(\varepsilon) \\
&\sim \int d\varepsilon \log|\varepsilon| \frac{\beta}{4 \cosh^2(\beta\varepsilon/2)} \\
&= \int dx \frac{1}{\beta} \log\left(\frac{|x|}{\beta}\right) \frac{\beta}{4 \cosh^2(x/2)} \\
&= -\log \beta \int dx \dots + \int dx \dots
\end{aligned} \tag{45}$$

so we get a

$$\chi_F \sim \log(1/T) \tag{46}$$

divergence as $T \rightarrow 0$.

- Antiferromagnetic: We differentiate

$$\begin{aligned}
\frac{\partial}{\partial \beta} \chi_{AF} &= \frac{\hbar^2}{4} \int d\varepsilon N_{\text{sing}}(\varepsilon) \frac{1}{\varepsilon} \frac{\partial}{\partial \beta} \tanh\left(\frac{\beta\varepsilon}{2}\right) \\
&= \frac{\hbar^2}{4} \int d\varepsilon A \log|\varepsilon| \frac{1}{\varepsilon} \frac{\varepsilon}{2} \frac{1}{\cosh^2 \frac{\beta\varepsilon}{2}} \\
&\sim \int dx \frac{1}{\beta} \log\left(\frac{|x|}{\beta}\right) \frac{1}{\cosh^2 \frac{x}{2}} \\
&= -\frac{\log \beta}{\beta} \underbrace{\int dx \dots}_{C_1} + \frac{1}{\beta} \underbrace{\int dx \dots}_{C_2}
\end{aligned} \tag{47}$$

and integrate

$$\begin{aligned}
\chi_{AF} &\sim \int d\beta \left[-\frac{\log \beta}{\beta} C_1 + \frac{1}{\beta} C_2 \right] \\
&= -\frac{1}{2} C_1 \log^2 \beta + C_2 \log \beta
\end{aligned} \tag{48}$$

so the main divergence is

$$\chi_{AF} \sim \log^2(1/T) \tag{49}$$

as $T \rightarrow 0$.

c)

• PROBLEM

Discuss how the results above are modified in $d \geq 3$ dimensions.

In $d \geq 3$, the DOS itself is finite at $\varepsilon = 0$. Therefore χ_F remains regular as $T \rightarrow 0$.

However, at half filling the dispersion is perfectly nested, $\varepsilon_{k+Q} = -\varepsilon_k$ for $Q = (\pi, \pi, \dots)$. Thus χ_{AF} still diverges, but only logarithmically,

$$\chi_{AF} \sim \log\left(\frac{1}{T}\right), \quad (50)$$

provided $N(0) \neq 0$. The stronger $\log^2(\frac{1}{T})$ divergence in $d = 2$ occurs because perfect nesting coincides with the logarithmic Van Hove singularity of the DOS.

d)

Consider now non-interacting electrons on a one-dimensional lattice with dispersion $\varepsilon_k = -2t \cos(ka)$ at half-filling ($\mu = 0$). Is there a Q -point in the Brillouin zone, $Q \in [0, 2\pi]$, for which $\varepsilon_{k+Q} = -\varepsilon_k = 0$? What is the signature of this “nesting” property in the free (bubble) susceptibility $\chi_0(Q, \omega = 0)$ (calculated in Exercise 2, Problem 4c)? Remember that the sum over k , $\sum_k$, can be replaced by $\int d\varepsilon N(\varepsilon)$ with the density of states $N(\varepsilon)$ from Exercise 1.

From Exercise 1, the one-dimensional nearest-neighbor tight-binding dispersion is

$$\varepsilon_k = -2t \cos(ka). \quad (51)$$

At half filling, the Fermi energy is

$$\varepsilon_F = 0, \quad (52)$$

and for $a = 1$ the Fermi points are

$$k = \pm \frac{\pi}{2}. \quad (53)$$

The corresponding one-dimensional DOS from Exercise 1 is

$$N_1(\varepsilon) = \begin{cases} \frac{1}{\pi\sqrt{4t^2 - \varepsilon^2}} & \text{if } |\varepsilon| < 2t \\ 0 & \text{if } |\varepsilon| > 2t \end{cases}. \quad (54)$$